

Bill of Materials

Urine Analyzer Lite | India Sourcing | Prices in INR (approx. ₹84 = \$1 USD)

Ref	Component	Description	Qty	Unit (₹)	Total (₹)	India Supplier	Sourcing Note
U1	ESP32-S3-DevKitC-1	Dual-core MCU with native USB; 8MB flash + 8MB OPI PSRAM	1	1299	1299	Robocraze	Also on Amazon.in ~₹1350; Evelta stocks too
U2	TP4056 USB-C Module	LiPo charging module with USB-C input and protection circuit	1	65	65	Robocraze	Very common; also at Probots / Amazon.in ~₹60
U3	S13V25F5 Buck-Boost Regulator	5 V 2.5 A step-up/step-down regulator from LiPo	1	1099	1099	Mouser India	Pololu product; Mouser.in stocks it. element14.in alternate
BT1	3.7V 2000mAh LiPo	Main rechargeable power source	1	599	599	Robocraze	Probots also carries; verify protection circuit included
U4	TCS34725 RGB Sensor #1	I2C RGB+clear color sensor with IR cut filter (primary pad sensor)	1	649	649	Robocraze	Evelta ~₹700; confirm IR-cut filter present on module
U5	TCS34725 RGB Sensor #2	I2C RGB+clear color sensor (secondary / differential)	1	649	649	Robocraze	Same part as U4; buy 2 together for small discount
U6	SHT31-D Module	High-accuracy I2C temperature and humidity sensor	1	749	749	Probots	Robocraze alternate ~₹780; avoid clone modules for accuracy
U7	BH1750 FVI	I2C digital ambient light sensor	1	149	149	Robocraze	Amazon.in ~₹160; very common; GY-302 breakout widely available
U8	ADS1115 ADC Module	16-bit I2C ADC with PGA for NTC thermistor	1	449	449	Robocraze	Probots ~₹400; Evelta ~₹480
RT1	NTC Thermistor 10k B3950	Temperature sensing inside optical chamber	1	25	25	Electronicscomp	Sunrom.com ~₹20; sold in packs of 5 — buy pack
U9	MX1508 Dual H-Bridge	Dual DC motor driver 2A per channel	1	120	120	Robocraze	Amazon.in ~₹100; Robokits ~₹110
M1	GA12-N20 6V 200RPM + Encoder	Compact DC gear motor with Hall-effect quadrature encoder	1	449	449	Robocraze	Robokits ~₹500; confirm 6V 200RPM variant with encoder
SW1	KW12-3 Micro Limit Switch	Home position limit switch (NO)	1	65	65	Robocraze	Amazon.in ₹50–80; buy pack of 5 for spares
SW2	KW12-3 Micro Limit Switch	End-of-travel limit switch (NO)	1	65	65	Robocraze	Same as SW1; order 4 total (2 working + 2 spares)
D1	High-CRI 5mm LED 95+ CRI	High colour-rendering white illumination LED for pad reading	1	220	220	Mouser India	IMPORT — no Indian-stocked 95+CRI 5mm LED equivalent; Nichia NSPWR70CSS-K1 via Mouser.in. Budget alternative: 90CRI Cree-equiv from Amazon.in ~₹80
Q1	2N7000 N-Channel MOSFET	PWM control MOSFET for illumination LED	1	15	15	Electronicscomp	Sunrom.com ~₹20; sold loose — buy 10 for <₹100
D2	275nm UVC LED Module	UV-C sterilization LED for tray decontamination	1	1050	1050	Mouser India	IMPORT — Mouser.in stocks Klaran / SETi UVC LEDs; order with heatsink. element14.in alternate. Allow 2-week lead time
U10	HX711 Load Cell Amplifier Kit 1kg	24-bit ADC for load cell weight measurement	1	320	320	Robocraze	Includes 1 kg bar-type load cell; Probots ~₹350

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PR1	CSN-A2 Micro Thermal Printer	58mm thermal receipt printer for diagnostic reports	1	2800	2800	Robocraze	Amazon.in ■2500–3200 depending on seller; verify TTL UART version (not USB-only)
LCD1	Waveshare 4.3in DSI LCD	Capacitive touch display; MIPI DSI interface	1	3999	3999	Evelta	Robocraze ~■4200; verify DSI cable included. Waveshare India authorised reseller: robu.in
SW3	Momentary Tactile Button 6x6mm	Scan trigger button	1	5	5	Electronicscomp	Sunrom ~■4; buy strip of 20 (<■80 total)
--	GT2 Timing Belt 200mm 6mm wide	Belt drive for strip-tray linear motion	1	199	199	Robocraze	Amazon.in ~■180; buy 2 (one spare)
--	GT2 Pulley 16T 3mm bore	Aluminium motor and idler pulleys	2	175	350	Robocraze	Amazon.in ~■150 each
--	M3 Stainless Socket Head Screws M3×10	General chassis fasteners	20	4	80	Amazon India	Buy 50-pack ~■150 for spares
--	M3 Brass Heat-Set Inserts M3×5	Threaded inserts for 3D-printed parts	12	8	96	Amazon India	Robocraze carries too; buy 20-pack
--	Frosted Acrylic Sheet 50×20×2mm	LED diffuser plate in optical chamber	1	180	180	Amazon India	Cut from A4 sheet ~■300; local acrylic shop in most cities cheaper
--	2mm Clear Acrylic Sheet (laser cut)	Strip flattening guard over tray	1	250	250	Amazon India	Local laser cutting shop (Lamington Rd Mumbai / Nehru Place Delhi) cheaper than online
--	Aluminium Sheet 80×40×2mm	UV-C safety shield (internal)	1	320	320	Amazon India	Local hardware/metal supplier; cut from 200×200mm sheet
--	Spectralon 99% White Standard 10×10mm	White calibration standard for sensor normalisation	1	2200	2200	Labsphere (import)	IMPORT — no India distributor; ship via FedEx International. DIY alternative: pressed BaSO4 tablet (~■200 at lab chemicals supplier like HiMedia)
--	Whatman Grade 1 Filter Paper Strips	Disposable tray liners pack of 100	100	12	1200	Merck India	HiMedia Laboratories (Mumbai/Delhi/Chennai) alternate; also Amazon.in ~■1400 for pack
--	Chassis Base (3D print – PLA)	Compact main chassis 160×120×55mm	1	300	300	Self / Local bureau	PLA ~■900/kg on Amazon.in (eSun / Bambu); 3D printing bureaus in major cities ~■5–8/gram
--	Optical Chamber (3D print – Black PLA)	Light-tight sensor housing 65×30×45mm	1	175	175	Self / Local bureau	Must be Black PLA; 100% infill adds material cost
--	Dipstick Tray (3D print – PLA)	Single 6mm strip channel with filter-paper slot	1	175	175	Self / Local bureau	Print in natural/white PLA; replace each batch run if contaminated
--	MCU Tray (3D print – PLA)	Snap-fit bracket for ESP32-S3-DevKitC-1	1	45	45	Self / Local bureau	Simple flat part; low print cost
--	Printer Bracket (3D print – PETG)	Mount for CSN-A2 thermal printer	1	220	220	Self / Local bureau	PETG ~■1200/kg; use for heat-resistant parts near printer
--	Display Bezel (3D print – PLA)	Front panel with 4.3in LCD cutout	1	130	130	Self / Local bureau	Sand and paint for professional finish
--	UV-C Shield (3D print – Black PLA)	Internal shroud for UV-C LED	1	90	90	Self / Local bureau	100% infill; UV-stable PLA preferred
--	Belt Tensioner (3D print – PLA)	Adjustable GT2 idler arm	1	90	90	Self / Local bureau	Pair with 625ZZ bearing (■60 on Amazon.in)
--	Limit Switch Bracket ×2 (3D print – PLA)	Adjustable KW12-3 mount	2	25	50	Self / Local bureau	Small part; minimal material

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Ref	Component	Description	Qty	Unit (■)	Total (■)	India Supplier	Sourcing Note
--	N20 Motor Bracket (3D print – PLA)	Adapter to mount N20 motor to chassis	1	45	45	Self / Local bureau	Print in PETG if near motor heat
				TOTAL (INR)		26156	
				TOTAL (USD approx @ ■84)		~311	

Note: Items marked IMPORT require ordering via Mouser India (mouser.in) or element14 India (in.element14.com). Allow 1–2 week lead time. 3D-printed parts use self-sourced PLA/PETG filament (~■900–1200/kg on Amazon.in).

Urine Analyzer Lite — Overview

Setup, calibration, firmware, and operation guide

Compact, automated urine dipstick reader by **Levram Lifesciences**.

Single-strip, N20-motor-driven, ESP32-S3 based with thermal print output.

Overview

The Urine Analyzer Lite reads standard 10-pad reagent dipsticks using dual TCS34725 RGB color sensors and classifies results for 10 analytes. A compact N20 gear motor steps the strip under the optical head at controlled speed. Results print via an integrated CSN-A2 thermal printer.

Key specs:

Parameter	Value
MCU	ESP32-S3-DevKitC-1 (Xtensa dual-core 240 MHz, 8 MB flash)
Sensors	TCS34725 ×2, SHT31, BH1750, ADS1115, HX711
Display	Waveshare 4.3" MIPI DSI capacitive touch LCD
Motor	GA12-N20 6V 200RPM with Hall encoder + MX1508 H-bridge
Printer	CSN-A2 58mm thermal (ESC/POS)
Power	3.7V 2000mAh LiPo → TP4056 USB-C charger → S13V25F5 5V regulator
Sterilization	275nm UV-C LED (10s cycle post-scan)
Analytes	Leukocytes, Nitrites, Urobilinogen, Protein, pH, Blood, Specific Gravity, Ketones, Bilirubin, Glucose
Estimated BOM	~\$387 per unit (prototype); target <\$250 at 50-unit volume

Repository Structure

```
urine-analyzer-lite/  
  ■■■ hardware/  
    ■ ■■■ schematic/  
      ■ ■ ■■■ gen_schematic.py # KiCad schematic generator  
      ■ ■ ■■■ urine_analyzer_lite.kicad_sch  
      ■ ■■■ cad/  
        ■ ■■■ main_chassis.scad # OpenSCAD enclosure (all 3D-printed parts)  
    ■■■ firmware/  
      ■ ■■■ main/  
        ■ ■■■ main.ino # Arduino sketch (state machine)  
        ■ ■■■ config.h # Pin map and tunable constants  
        ■ ■■■ sensors.h # All sensor drivers  
        ■ ■■■ motor.h # N20 motion control + homing  
        ■ ■■■ analysis.h # HSV pad classification (10 analytes)
```

```
■ ■■■ printer.h # ESC/POS thermal print
■ ■■■ bom/
■ ■■■ bom.csv # Full bill of materials (~40 items)
■ ■■■ TODO.md # Project checklist
■ ■■■ README.md
```

Hardware

Schematic

Open in KiCad 7+:

```
File → Open → hardware/schematic/urine_analyzer_lite.kicad_sch
```

To regenerate after edits:

```
cd hardware/schematic && python3 gen_schematic.py
```

CAD

Open in OpenSCAD ≥ 2021.01:

```
hardware/cad/main_chassis.scad
```

All 3D-printable parts are exported from one file. Render individual modules:

```
openscad -o chassis_base.stl main_chassis.scad
```

Print settings:

Part	Material	Infill	Layer
Chassis base, MCU tray, dipstick tray	PLA	20%	0.2 mm
Optical chamber, UV-C shield	Black PLA	100%	0.1 mm
Sensor mounts, printer bracket, belt parts	PETG	30%	0.2 mm

Firmware

Requirements

- Arduino IDE ≥ 2.3 with ESP32 board package ≥ 2.0.14
- Board: **ESP32S3 Dev Module** — Flash: 8MB, PSRAM: OPI, USB: Hardware CDC

Libraries (Arduino Library Manager)

```
Adafruit TCS34725 >= 1.4.2
Adafruit ADS1X15 >= 2.5.0
Adafruit SHT31 >= 2.2.2
BH1750 by Christopher Laws >= 1.3.0
HX711 by bogde >= 0.7.5
```

Flash

1. Open `firmware/main/main.ino` in Arduino IDE
2. Select board and port
3. Upload

Calibration (required before first use)

1. **White standard** — place Spectralon pad under optical head; record R/G/B values in `config.h` (`CAL_R/G/B_WHITE`)
2. **Scale** — place known weight on scale platform; adjust `scale.set_scale()` factor in `sensors.h`
3. **Motor steps** — measure real pad pitch with calipers; adjust `TICKS_PER_PAD` in `config.h`
4. **Hue ranges** — run known positive/negative strips; refine `PAD_TABLE[]` in `analysis.h`

Operation

1. Insert dipstick tray with strip face-up
2. Place sample cup on weight sensor (minimum 1 g detected)
3. Press scan button
4. Device homes, steps across all 10 pads, classifies, prints report
5. UV-C sterilization cycle runs automatically (10 s)
6. Remove strip; device returns to IDLE

Bill of Materials

See `bom/bom.csv` — estimated prototype total: **\$387 USD**.

License

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For clinical use by qualified personnel only.

Hardware · Firmware · Procurement · Integration · Regulatory

Hardware — PCB / Schematic

- ## Hardware — CAD / Mechanical

- ## Firmware

- ✓ `config.h` — all pin assignments and tunable constants
 - ✓ `sensors.h` — TCS34725, ADS1115, SHT31, BH1750, HX711 drivers
 - ✓ `motor.h` — N20 encoder-based motion control + homing
 - ✓ `analysis.h` — HSV-based pad classification (10 analytes)
 - ✓ `printer.h` — CSN-A2 ESC/POS thermal printer output
 - ✓ `main.ino` — state machine: IDLE → HOMING → SCANNING → PRINTING → STERILIZING
 - ■ Flash firmware to hardware and test boot
 - ■ Calibrate HX711 scale factor with known weight
 - ■ Calibrate TCS34725 against Spectralon white standard (update `CAL_R/G/B_WHITE`)
 - ■ Calibrate `TICKS_PER_PAD` with physical dipstick (measure real pad pitch)
 - ■ Validate HSV hue ranges in `analysis.h` using positive/negative reference strips
 - ■ Implement display UI (LVGL or LovyanGFX on DSI LCD)
 - ■ Add Bluetooth/Wi-Fi result upload (optional)
 - ■ Implement OTA firmware update
 - ■ Add watchdog timer and brownout detection
 - ■ Write unit tests for `rgb_to_hsv()` and `classify_pad()`
-

BOM / Procurement

- ✓ BOM CSV (`bom/bom.csv`) — 40 line items, estimated total ≈ \$387 per unit
 - ■ Obtain quotes from Indian distributors (Rhydo, Robocraze, Electronicscomp)
 - ■ Source TCS34725 from Adafruit or equivalent (check availability)
 - ■ Procure Labsphere Spectralon calibration standard (long lead time)
 - ■ Verify CSN-A2 printer stock and 58mm paper availability locally
 - ■ Confirm Waveshare 4.3in DSI LCD DSI compatibility with ESP32-S3
 - ■ Bulk pricing target: < \$250 per unit at 50-unit volume
-

Integration & Testing

- ■ Power-on test: verify 3V3 and 5V rails under load
 - ■ I2C scan — confirm all 5 devices at expected addresses
 - ■ Motor homing test — both limit switches trigger correctly
 - ■ Full scan cycle with blank dipstick — no crashes, 10 pads detected
 - ■ Full scan cycle with positive control strip — correct analyte results
 - ■ Thermal print output matches on-screen report
 - ■ UV-C sterilization cycle runs for 10 s and shuts off
 - ■ Weight sensor detects cup placement > 1 g threshold
 - ■ Environmental envelope test: 15–35 °C, 20–80% RH
-

Documentation

- ✓ Assembly guide template (hardware/schematic/ + GUIDE.md in source data)
 - ■ Write complete assembly guide with photos
 - ■ Write calibration SOP (white standard → pad validation)
 - ■ Write operator quick-start card (print on 58mm paper)
 - ■ Add troubleshooting section (I2C errors, motor faults, print jams)
 - ■ Create block diagram image for README
-

Regulatory / Quality (future)

- ■ CE / BIS marking assessment
- ■ IEC 61010-1 electrical safety review
- ■ ISO 13485 QMS alignment checklist
- ■ 510(k) pathway assessment (if exporting to US)